WENDELL LEÃO

Senior Unity Developer

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PROFESSIONAL SUMMARY

5+ years of experience building gameplay systems, multiplayer features, and scalable architecture for AA PC and mobile titles. Strong in architecture, refactoring legacy code, and implementing complex data-driven systems, connecting front end to backend and integrating services. Known for resolving critical bugs under tight deadlines, clear communication, and workflow improvements, with AI-assisted automation, documentation, and unit testing built into daily work.

SKILLS

- **Architecture:** SOLID Principles, Dependency Injection, State Machines, Factory, Command, Observer.
- **Programming:** C#, Gameplay Systems, Engine Architecture, Scriptable Objects, Client-Side Prediction.
- **Unity:** Animator/Mecanim, StateMachineBehaviours, Cinemachine, Splines, URP/HDRP, iOS/Android.
- **Performance:** CPU/GPU Profiling, Addressables, Asset Bundles, Object Pooling, Async Programming, UniTask.
- **Multiplayer:** FishNet, Mirror, Photon Fusion/Quantum, Netcode for GameObjects, UGS, PlayFab, PlayFlow.
- **User Interface:** UI Toolkit, uGUI, Virtualized Lists, Data Driven UI Architecture.
- **Backend & Servers:** Linux Dedicated Servers, PM2, Node.js, WebSockets, Custom UDP Layer, REST APIs.
- **Tools:** Steamworks API, Git/Git Flow, CI/CD, Editor Tooling/Automation, Opsive UCC, Claude Code, Unity MCP.
- **Workflow:** Agile/Scrum, GitHub, GitLab, Asana, Notion, Figma, Slack, Obsidian, pCloud.

WORK EXPERIENCE

Senior Unity Developer - BGNS Studios - England **May 2024 - Present**

The Platform on Steam

- Joined an **8-programmer team** building a user-generated content platform in the style of Roblox and Rec Room, designed for **100 concurrent players**. Brought in as a senior reinforcement to resolve critical stability issues across PC and VR ahead of a Gamescom (Germany) showcase.
- Diagnosed and fixed a critical, **100% reproducible** late-joiner crash in the FishNet networking wrapper, tracing it to incorrectly cached server-replicated objects that produced duplicate object IDs.
- Implemented **server-side prediction and reconciliation** for player movement using FishNet, integrating it into Opsive's Ultimate Character Controller (UCC) through partial classes and helpers that kept the networking layer separate from UCC's internal code, preserving compatibility with future package updates. Extended it to flight, aiming, moving platforms, knockback, and respawn, skipping replays when server state matches local history and fixing platform jitter and authority handoff races.
- Deployed and managed dedicated Linux servers via PlayFlow Cloud to support multiplayer testing across first and third-person minigames, debugging server-only issues through server logs and pure-client reproductions.
- Owned the continuous refinement of the **in-game AI coding feature**, where players describe behavior in natural language and an LLM generates the JavaScript that runs on game events: tuned prompts and command documentation so the model picks the right commands, and built the Unity side (per-event code editing, reset-to-baseline for items and NPCs, script lifecycle fixes).
- Built and maintained **15+ commands** on the WebSocket command bus connecting the Unity client, the FishNet dedicated server, and the Node.js backend, owning each one from Unity handler to backend catalog entry.
- Built an AI-assisted development workflow: authored custom Claude Code skills for commit conventions, architecture, code style, and engine guidelines, maintain an Obsidian vault of **180+ notes** as persistent agent memory to cut token usage, and wrote **69 NUnit EditMode and 13 Node tests** with AI agents under my review.

PIX Football Manager on Steam

- Implemented RESTful API integration in Unity via Swagger UI to synchronize player data and contract life cycles, and built **diegetic UI Toolkit interfaces**, revamping legacy screens into an intuitive layout.
- Developed and owned the team creation and customization workflow, engineering a complex emblem design system (layered composition, shapes, patterns, and colors) alongside the training facility interface using UI Toolkit.

Draft Fever Bowl on Steam

- Served as the frontend technical representative for an AA PC title, representing a four-developer frontend team in cross-department planning, stakeholder presentations, and architectural decisions, and led onboarding for 4 incoming frontend programmers (game scope, Git Flow, codebase, core technologies) until full ramp up.
- Architected large-scale scouting and drafting systems using C#, Scriptable Objects, and Virtualized Lists, supporting over **2,000** real-time synchronized entities while increasing critical screen performance from **40 FPS to 80-100 FPS**. Also built the interactive 3D scouting map and the multiplayer draft event.
- Directed a full Scriptable Render Pipeline migration (HDRP to URP), refactoring rendering-dependent systems and custom shaders boosting menus to **80 FPS** and stabilizing gameplay from **20-30 FPS to a solid 60 FPS**.
- Owned and redesigned the runtime character generation pipeline (backend data, facial customization, blend shapes, automated portraits, asset caching), reducing load times from **4 minutes to approximately 50 seconds**.
- Identified rendering bottlenecks within runtime texture generation pipelines and implemented asynchronous processing using UniTask and semaphore-based synchronization, eliminating frame drops during live emblem creation and visual asset generation.

Mid-Level Unity Developer - Triplano Games - Brazil **Jan 2024 - May 2024**

Amazon Warriors on App Store

- Served as **sole programmer** on this mobile adaptation of a well known Brazilian children's franchise, with full autonomy over technical decisions while aligning scope with art, design, sound, and production.
- Built the complete UI framework (progression, unlocks, shop, customization), later adopted across other studio projects, and used object pooling to eliminate spikes in procedural terrain generation, alongside character abilities, enemy AI, backend integration, monetization, and analytics, maintaining stable **30 FPS** on mobile.

Mid-Level Unity Developer - Advance Garde - Brazil **Dec 2021 - Dec 2023**

Rogue Masters on Steam

- Managed and expanded a mature multiplayer architecture using Mirror (P2P) for a 6-player AA Steam title, adding Steamworks integration (room browsers, filtering, passcode-protected sessions, and Steam invitations).
- Resolved a **critical memory leak** in the Mirror networking stack that caused progressive crashes in long sessions, using cross-version analysis to find the root cause and integrate a targeted upgrade path prior to launch.
- Resolved legacy engine-freezing issues by implementing an asynchronous bootstrap/startup architecture for Steamworks, backend services, and SDK initialization, reducing startup times by **approximately 60%** and significantly improving game stability.
- Engineered server/client synchronization for high-paced action mechanics, coupling custom network state machines with Unity's Animator (Mecanim) and StateMachineBehaviours to replicate movesets, AI behaviors, VFX, and buff/debuff states.
- Redesigned the **character-assembly pipeline**, replacing expensive synchronization with lightweight ID-driven updates for weapons, armor, movesets, and enchantments, eliminating equipment-change frame drops.
- Developed multiplayer UI systems including scoreboards, end-match summaries, MVP calculations, voting, spectator mode, and HUD elements synchronized across networked clients.

Junior Unity Developer - Main Leaf Games - Brazil **Aug 2021 - Dec 2021**

Pet Shop Fever on Google Play

- Contributed to a title that surpassed **1M downloads**, built under strict guidelines and detailed code review, applying the studio's proprietary SDK patterns (save systems, service locator, object pooling).
- Architected an additive scene UI framework with dynamic UI streaming to minimize memory footprint on low end devices, and reduced build size and draw calls through texture atlasing, Addressables, and Asset Bundles.

LANGUAGES

Portuguese: Native | **English:** Professional proficiency (advanced)

EDUCATION

Technology Degree in Digital Game Development - FMU, Brazil (Feb 2019 - July 2021)